Revision Notes for Class 7 Maths Chapter 1 – Large Numbers Around Us

India is a land where large numbers often come up in daily life. For example, people talk about 'one lakh varieties of rice' or the population of a city being more than a lakh. Understanding large numbers is an important part of everyday maths, especially when dealing with populations, heights, or big calculations. In this chapter, students explore the world of large numbers and see how to read, write, compare, round, and use them in real-life examples.

The Meaning and Scale of Large Numbers

A lakh is a familiar term in India and is written as 1,00,000. It is also called "one hundred thousand" in the international system. To get a ballpark idea: if someone ate one new variety of rice each day, they would still need over 270 years to try a lakh varieties! This illustrates just how vast a lakh is, especially in the context of time, length, or population. Similarly, bigger figures like crores (1,00,00,000) and arabs (1,00,00,00,000) make an appearance when we deal with national statistics or large companies.

Understanding Place Value Systems

There are two main systems for reading and writing large numbers: the Indian place value system and the International (American) system. In the Indian system, commas are placed according to the groups – starting from the right: three digits (thousand), then each group of two digits (lakhs, crores, arabs). For example, 12,34,56,789 is read as twelve crore thirty-four lakh fifty-six thousand seven hundred eighty-nine. In the American system, commas are placed in groups of three: million, billion, trillion. For instance, 123,456,789 = one hundred twenty-three million, four hundred fifty-six thousand, seven hundred eighty-nine.

Comparing and Relating Big Numbers

When trying to get a sense of how big something is, it helps to compare it to something more familiar. The heights of famous structures like the Statue of Unity (about 180 meters) or Kunchikal waterfall (about 450 meters) are used to relate to everyday buildings, such as calculating how many floors a building would need to reach similar heights. Similarly, the length a lakh people stretched shoulder-to-shoulder forms a line of about 38 kilometers!

Is a Lakh Big or Small?

Whether one lakh is "big" or "small" depends on context. It is large when talking about types of rice or living 1 lakh days (about 274 years!). But when we see a cricket stadium with a lakh spectators, or count the hundred thousand hairs on a person's head, it doesn't seem so huge. This shows that large numbers can seem big in some instances and not in others.

Reading and Writing Large Numbers

Commas are extremely helpful in breaking up numbers for easier reading. In the Indian system, 45,830 (forty-five thousand eight hundred thirty) uses a comma after the thousands place. The number 12,78,830 is written as "twelve lakh seventy-eight thousand eight hundred thirty." Practice is essential for converting between number names and numerals in both systems, and for writing numbers in words up to crores and beyond.

Fun With "Calculator Buttons"

There are creative activities in this chapter that help students understand large numbers through calculators with different button values: +1, +10, +100, +1,000, etc. For example, to reach 1,00,000 using only a +1,000 button, you would press it 100 times. Exploring numbers with such limitations helps students break down numbers into groups of tens, hundreds, or thousands, and builds a strong sense of place value.

Converting Between Systems

A helpful table shows how to convert between the Indian and American systems:

Indian System	American System

1,000 (One thousand)	1,000 (One thousand)
10,000 (Ten thousand)	10,000 (Ten thousand)
1,00,000 (One lakh)	100,000 (Hundred thousand)
10,00,000 (Ten lakh)	1,000,000 (One million)
1,00,00,000 (One crore)	10,000,000 (Ten million)
1,00,00,00,000 (One arab)	1,000,000,000 (One billion)

The Indian system is widely used in South Asian countries, with words 'lakh' and 'crore' coming from Sanskrit. In the American system, groups like million, billion, trillion are based on thousands.

Rounding and Approximation

Often, we use rounded or approximate values in daily life because exact numbers aren't always needed. For example, reporting a population as "about 75,000" gives a clearer idea than using the precise number 76,068. We round up when we want to ensure enough of an item, and round down when we want a safer, smaller estimate.

Using Nearest Neighbours

Finding the nearest thousand, ten thousand, lakh, or crore to a given number helps in estimation. For example, for 6,72,85,183, the nearest crore would be 7,00,00,000. This technique is especially helpful for quick calculations, estimation, and population studies.

Multiplication Tricks and Patterns

Multiplying by 10, 100 and their multiples is made easier using shortcuts. For instance, 116 × 5 can be solved as (116 × 10)/2. Similarly, multiplying by 25 is like multiplying by 100 and dividing by 4. Recognising these patterns helps with calculations involving large numbers quickly and efficiently. When two 2-digit numbers are multiplied, the result is always a 3- or 4-digit number.

Real-Life Examples With Large Numbers

Tables in the chapter show population statistics of Indian cities (e.g., Mumbai: over 1.24 crore in 2011), puzzles comparing population changes over time, and interesting facts such as the number of bacteria in soil and the water released by Amazon river each second. These real-life connections make the concepts lively and relevant.

Fun Thought Experiments and Puzzles

Students are encouraged to solve puzzles, such as: Could all of Mumbai's population fit in one lakh buses? If a paper weighs 5g, could you lift one lakh sheets? How far could you travel in ten years at 100km a day? These activities help in developing reasoning with big numbers.

Summary of Key Points

- A lakh is written as 1,00,000 (one hundred thousand in international system), and a crore as 1,00,00,000 (ten million).
- Large numbers are made more understandable using place value, rounding, and real-life comparisons.
- Patterns and shortcuts simplify arithmetic with big numbers.
- Context matters—a number can feel "big" or "small" based on what is being measured.
- Estimation and rounding are practical tools for reports and calculations.

Creative Activities With Digits

The chapter ends with playfully building numbers with matchsticks, rearranging digits, and forming puzzles, strengthening the basics of how large and small numbers are constructed and visualized.